Redshift in FFGFT: static, achromatic

and the z →time translation onto the expanding picture
with the cosmological degeneracy (Doc. 267)

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Contents

What this is about

In FFGFT the cosmological redshift arises *not* from spatial expansion but from a static geometric mechanism in the ξ field. This note fixes three points and corrects a fourth:

1. **Static:** $1+z=e^{\xi x}$, hence $z \approx \xi x$ – energy loss / fractal path-lengthening along the light path, no scale factor $a(t)$ (Doc. 064, 025/026).
2. **Achromatic:** the fractal path-lengthening is purely geometric and stretches *all* wavelengths by the same factor – it is *not* wavelength-dependent. Finite-element simulations of the geometry confirm this. (**Correction** of the earlier chromatic presentation; Doc. 190 C6.)
3. **The $z\rightarrow$time axis is a ΛCDM construct:** because FFGFT has no $a(t)$, plotting any quantity "over cosmic time" (e.g. the Lilly–Madau star-formation rate) is only a *translation of the observation onto the expanding picture*.
4. **The data fit – via the cosmological degeneracy** (Doc. 267): FFGFT reproduces the same observables as ΛCDM, with no false prediction, but also no single decisive test.

Throughout: $\xi = \frac{4}{30000} = 1{,}3333\times 10^{-4}$, $H_0^{\text{T0}} \approx 66{,}2$ km/s/Mpc.

.1 The static redshift

In the static universe a photon loses energy along the light path according to

$$\frac{dE}{dx} = -\xi E \qquad \Longrightarrow \qquad E(x) = E_0\, e^{-\xi x}.$$

The linear dependence on E is decisive: it makes the *leading* effect frequency-independent. The redshift is

$$z = \frac{E_0 - E(x)}{E(x)} = e^{\xi x} - 1, \qquad \boxed{1+z = e^{\xi x}}, \qquad z \approx \xi x \text{ (for small } z).$$

Comparison with the Hubble law $z = H_0 d/c$ gives (natural units, $c=1$) the identification

$$H_0^{\text{T0}} = E_H/\hbar, \qquad E_H = E_0\,\xi^{41/4} = 1{,}41\times 10^{-33}\ \text{eV} \;\Rightarrow\; H_0^{\text{T0}} \approx 66{,}2\ \text{km/s/Mpc}$$

(Doc. 064/026; the exponent $41/4$ is not derived from first principles, Doc. 190 R20). The *light-travel time* to an object at redshift z is therefore

$$\boxed{\tau(z) = \frac{\ln(1+z)}{H_0^{\text{T0}}}}.$$

There is *no* finite cosmic age and *no* Big Bang: τ grows without bound as $\ln(1+z)$.

.2 Achromatic, not wavelength-dependent (correction)

Earlier documents (Doc. 004, 025, 030, 039, 041, 053, 061, 063, 068, 081) stated a *wavelength-dependent* redshift (e.g. $z(\lambda_0) \propto \lambda_0$, "UV stronger than radio"). That presentation is **superseded**.

The mature FFGFT mechanism is *fractal path-lengthening*: the path the signal travels through the recursive time geometry is stretched by a factor that depends only on distance/geometry – *not* on the photon wavelength. Hence $1+z=e^{\xi x}$ holds for *all* wavelengths with the same factor: the redshift is **achromatic**.

Evidence: finite-element simulations of the fractal path-lengthening yield *no* wavelength-dependent redshift. This is also physically compelling: an *energy-/wavelength-specific* loss term would be chromatic, but a *geometric* stretch factor acts on wavelength *and* pulse duration in the same ratio (as do gravitational and standard-cosmology redshift, also achromatic).

Consequence for observation: the cosmological redshift is achromatic to high precision – FFGFT therefore gets this *right*. The earlier "wavelength-dependent discriminator" drops out; it moves from the "test" column to the "fits" column. The correction is recorded in Doc. 190 as **C6**; the affected documents are listed there.

.3 The z →time mapping is model-dependent

Only the *redshift* z (and the quantity measured at that z) is model-independent. The conversion z →time, by contrast, is a modelling choice:

- Λ**CDM** computes it via $a(t)$ (Friedmann integral). The look-back time levels off at $t_0 \approx 13{,}8$ Gyr – the "Big Bang ceiling". That ceiling exists *only because expansion is assumed*.
- **FFGFT** computes it via $\tau = \ln(1+z)/H_0^{\text{T0}}$. There is no ceiling; τ grows without bound.

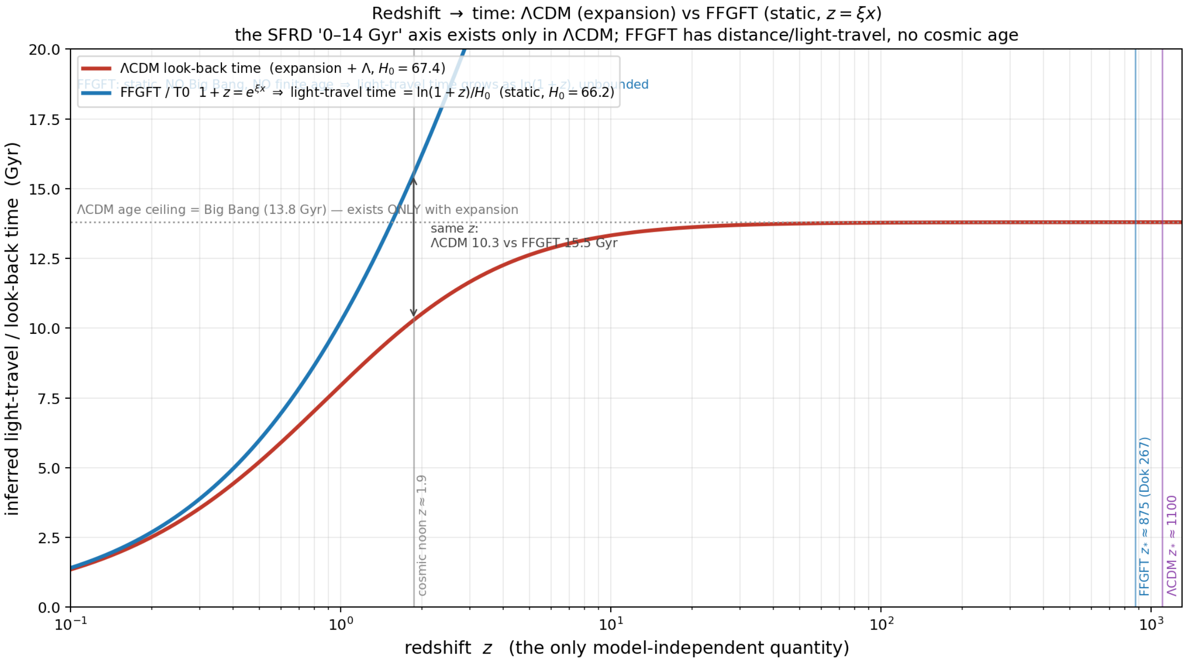

Figure 1: Redshift → time is model-dependent. ΛCDM look-back (red, expansion $+\Lambda$) saturates at the Big Bang ceiling $13{,}8$ Gyr; FFGFT (blue, static, $1+z=e^{\xi x}$) gives an unbounded light-travel time $\ln(1+z)/H_0$ with no Big Bang. At cosmic noon ($z \approx 1{,}9$): ΛCDM $10{,}3$ vs. FFGFT $15{,}5$ Gyr. Recombination: ΛCDM $z_* \approx 1100$, FFGFT $z_* \approx 875$ (Doc. 267).

Implication: plotting "star-formation rate over cosmic time 0–14 Gyr" (Lilly–Madau) is intrinsically a ΛCDM act. In FFGFT there is no cosmic time axis – only distance / light-travel time. The data points (z, SFRD) remain valid; placing them on a 0–14 Gyr axis *is* already the translation onto the expanding picture.

.4 How do the data fit? – the cosmological degeneracy

Doc. 267 establishes the central fact: FFGFT and ΛCDM are *observationally degenerate* on the standard tests. Whatever produces a given $z(d)$ relation produces the same fluxes, angles and time dilations.

- **Supernova time dilation:** distant SN Ia run stretched, $b \approx 1$ (DES: $b = 1{,}003 \pm 0{,}011$). The fractal path-lengthening stretches wavelength *and* pulse duration in the same ratio $\Rightarrow b = 1$. The test only excludes *non-dilating* tired-light ($b = 0$) – FFGFT predicts $b = 1$ and is unaffected.
- **BAO angular scales:** the non-linear temporal recursion curves $z(d)$ like an accelerated expansion $\Rightarrow$ the same angles. Caveat R20: BAO uses D_M/r_s with r_s from ΛCDM – model vs. model.
- **CMB peaks:** both reproduce the peak structure; ξ plays no role for the peaks; both equally circular (Doc. 267/268).
- **Lilly–Madau / SFRD:** the z-space signal (rise to $z \sim 2$, then fall) is model-independent and in FFGFT astrophysical galaxy evolution over light-travel time. The SFRD *values* ($M_\odot$/yr/Mpc^3), however, are doubly ΛCDM-derived – via the luminosity distance $d_L(z)$ and the comoving volume. A fair FFGFT test re-derives them with $d(z)$ and volume from $1 + z = e^{\xi x}$.

Honest status: the degeneracy cuts both ways. The data *fit* FFGFT (no false prediction, achromatic as observed), but they do not *compel* it. The case for FFGFT therefore rests on *parsimony* (no Λ, no dark energy, no H_0 tension – Doc. 064) and on deriving the constants from ξ – not on a single discriminator.

.5 No Big Bang – the direction of development reverses

FFGFT has no Big Bang. The apparent high-redshift "onset" – the fall of the SFRD and the supposed beginning of the first galaxies at $z \sim 14$ – is in FFGFT *not* a temporal beginning but a *geometric turning point* (Doc. 263).

The observed quantity is the cumulative fractal path-lengthening $z_{\text{obs}} = (L_{\text{eff}} - d)/d$ (Doc. 026/263). For small z it is linear, $z(d) = \kappa d$ with $\kappa = E_H/(\hbar c)$ – the slope ΛCDM names "H_0". Near the cosmological maximum scale R_H, however, the $z(d)$ curve has an *extremum*: FFGFT *computes* there $z_* \approx 875$ (Doc. 267, reference value; caveat R20) – the same surface ΛCDM reads as recombination/decoupling at $z \approx 1089$. The ΛCDM value 1089 is thus only the expansion reading; the FFGFT-own value is 875, *not* 1100. *Beyond* R_H one has $z \to \infty$ – with no particle horizon and no Big Bang.

What is z_* from the FFGFT view? It is *not* a recombination time. FFGFT has no hot early phase, no photon-baryon plasma, no decoupling. The CMB is the *stationary quantum ground state* of the Φ field (Doc. 025), and its acoustic peaks are discrete T^4 *lattice resonances* (analogous to the emission lines of an atom or phonon dispersions in a crystal) – *not* frozen sound waves of a primordial plasma. The numerical value itself is a *geometric ratio* (Doc. 267):

$$z_* \approx \left(\frac{\lambda_e}{L_0}\right)^{1/\ln(1/\xi)} \approx 875,$$

with the reduced Compton wavelength λ_e and the sub-Planck length $L_0 = \xi \ell_P$ – i.e. a *resonance / structure scale* of the T^4 lattice, not a thermal event in time. (Caveat R20: the exponent $1/\ln(1/\xi)$ is not yet derived from the T^4 geometry; z_* is currently a reference value of the right order of magnitude, not a derived FFGFT result.)

Thus **the apparent direction of development reverses at the maximum scale**: what ΛCDM reads as "earlier = faster development up to a beginning" is in FFGFT the approach to the geometric extremum near R_H. The high-redshift fall of the SFRD is therefore the approach to this turning point, *not* a beginning. H_0 is emergent, not a fundamental parameter – in the compact T^4 form there is no H_0 at all (Doc. 263).

Symmetrically open is only the forward derivation of the exponent $41/4$ from ξ (Doc. 190 R20) – missing for every framework, not only FFGFT.

Status

Static redshift $1 + z = e^{\xi x}$: *core* (Doc. 064/026). Achromaticity: *corrected/established* (FE simulation; Doc. 190 C6). $z \to$time as a ΛCDM translation and cosmological degeneracy: *established* (Doc. 267). Status throughout per Doc. 190.